Example 1.8 Write the following fractions as decimals.

(i) $\frac{2}{5}$ (ii) $\frac{3}{4}$ (iii) $\frac{9}{1000}$ (iv) $\frac{1}{50}$ (v) $3\frac{1}{5}$

Solution

(i) We have to find a fraction equivalent to $\frac{2}{5}$ whose denominator is 10.

$$\frac{2}{5}=\frac{2\times 2}{5\times 2}=\frac{4}{10}=0.4$$

(ii) We have to find a fraction equivalent to $\frac{3}{4}$ with denominator 100. Since there is no whole number that gives 10 when multiplied by 4, let us make the denominator as 100.

So, $\frac{3}{4}=\frac{3\times 25}{4\times 25}=\frac{75}{100}=0.75$

(iii) In $\frac{9}{1000}$, tenth and hundredth place is zero.

Therefore, $\frac{9}{1000}=0.009$

(iv) We have to find a fraction equivalent to $\frac{1}{50}$ whose denominator is 100.

$$\frac{1}{50}=\frac{1\times 2}{50\times 2}=\frac{2}{100}=0.02$$

(v) In $3\frac{1}{5}$, keep the whole number 3 as such, we can find a fraction equivalent to $\frac{1}{5}$ with denominator 10.

$$3+\frac{1}{5}=3+\frac{1\times 2}{5\times 2}=3+\frac{2}{10}=3.2$$

Example 1.9 Convert the following into simplest fractions.

(i) 0.04 (ii) 3.46 (iii) 0.862

Solution

(i) $0.04=\frac{4}{100}=\frac{1}{25}$

(ii) $3.46=3+\frac{46}{100}$

$$=3+\frac{46\div 2}{100\div 2}$$

$$=3+\frac{23}{50}$$

$$=3\frac{23}{50}$$

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